

STAREHE GIRLS' 2025 PREDICTION



POSSIBLE KCSE EXAM

233/3



CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

SCHOOL.....

SIGN.....

CONFIDENTIAL

INSTRUCTION TO SCHOOLS

The information contained in this paper is to enable the head of the school and the teacher in charge of chemistry to make adequate preparations for this year's chemistry practical Joint examination.

NO ONE ELSE should have access to this paper or acquire knowledge of its contents. Great care **must** be taken to ensure that the information herein does not reach the candidates either directly or indirectly. Chemistry teachers **SHOULD NOT** perform any of the experiments in the same room as the candidates or make the results of the experiments available to the candidates or give any other information related to the experiments.

In addition to the equipment, apparatus and chemicals in an ordinary Laboratory, each candidate require:

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|---|--|
| 1. About 150cm ³ of solution H. | 16.2 boiling tubes |
| 2. About 150cm ³ solution J | 17. Filter funnel |
| 3. About 80cm ³ solution K | 18. Test tube holder |
| 4. Burette | 19. 5 clean dry test tubes |
| 5. Pipette 25 ml | 20. 100 ml plastic beaker |
| 6. 1 pipette filler | 21. Metallic spatula |
| 7. 2 Conical flask | 22. About 0.5g solid U in a stoppered container. |
| 8. Means of labelling | 23. About 0.5g Solid P in a stoppered container. |
| 9. 250ml volumetric flask | 24. 1 label |
| 10. White tile | 25. About 0.5g of sodium hydrogen carbonate. |
| 11. 250ml plastic beaker | |
| 12. tissue paper | |
| 13. Thermometer (-10-110°C) | |
| 14. 50ml measuring cylinder | |
| 15. 500 ml Distilled water in a wash bottle | |

Access solutions

- ❖ 2M sodium hydroxide supplied with a dropper.
- ❖ 2M ammonia solution supplied with a dropper.
- ❖ 0.1M Lead (II) nitrate supplied with a dropper.
- ❖ 0.1M barium nitrate solution.
- ❖ 0.2M Potassium iodide.
- ❖ Acidified potassium manganate (VII)
- ❖ Acidified potassium dichromate (VI)
- ❖ Phenolphthalein indicator supplied with a dropper.
- ❖ Source of heat.

NOTES

1. 2M sodium hydroxide is prepared by dissolving 80 g of sodium hydroxide pellets in about 200 cm³ of distilled water and topping up to 1 litre of solution
2. 2M ammonia solution is prepared by diluting 112 ml of concentrated ammonia solution in about 200 cm³ of distilled water and top up to 1 litre.
3. 0.1 M lead (II) nitrate is prepared by dissolving 33.0 g of solid lead (II) nitrate in about 400 ml of distilled water and top up to 1 litre of solution.
4. 0.1 M barium nitrate is prepared by dissolving 26 g of solid lead (II) nitrate in about 400 cm³ of distilled water and top up to 1 litre of solution.
5. 0.2M Potassium iodide is prepared by dissolving 33g in about 400cm³ of water and dilute to one litre of solution.
6. Solid U is maleic acid.
7. Solid P is Zinc chloride.
8. Solution H is prepared by dissolving 55 ml of concentrated Sulphuric (VI) acid in about 600cm³ of distilled water and dilute to 1 litre of solution.
9. Solution J is prepared by dissolving 80 g of sodium hydroxide in 400 cm³ of distilled water and top up to one litre of solution using distilled water.
10. Solution K is prepared by dissolving 25g of citric acid C₃H₅O(COOH)₃ RFM=192 in about 400cm³ and dilute to 1 litre of solution. (0.13M Citric acid)
11. Acidified potassium manganate (VII) is prepared by dissolving 3.12g of potassium manganate (VII) in 200cm³ of 2M sulphuric (VI) acid and making it to one litre.
12. Acidified potassium dichromate (VI) is prepared by dissolving 25g of potassium dichromate (VI) in 400cm³ of 2M sulphuric (VI) acid and making it to one litre.